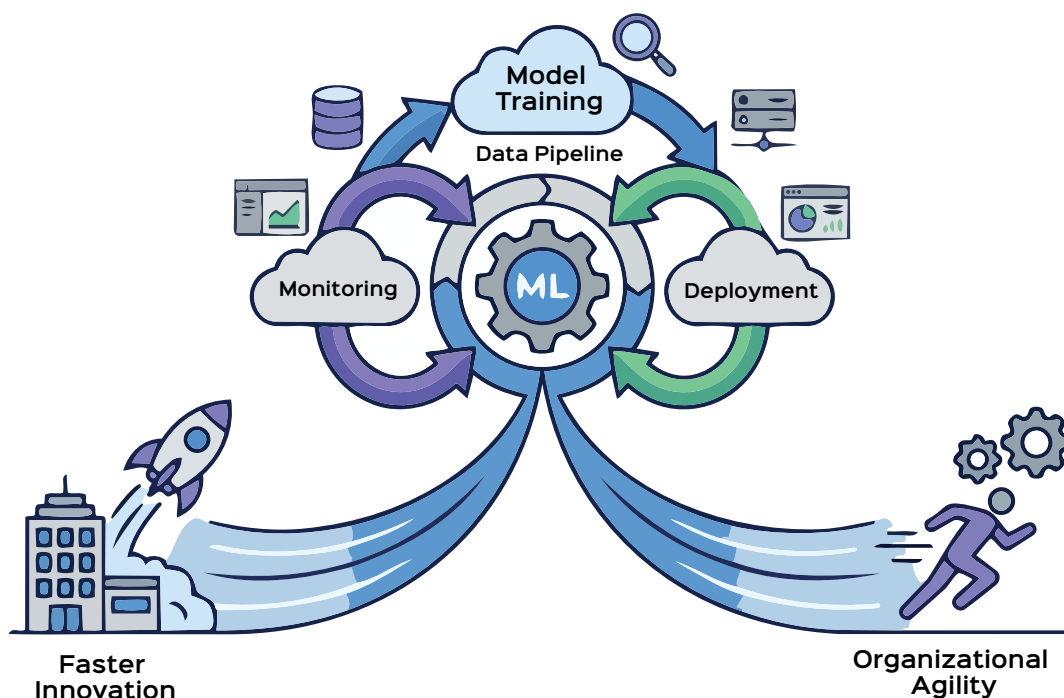


MLOps for Enterprise Agility: An Overview

MLOps helps enterprises develop, deploy, and manage machine learning systems that are reliable, scalable, and production ready.



Enterprises have moved past the era of siloed AI experiments and into a phase of true operational reliance. The goal is to automate and stabilise the entire lifecycle of a machine learning model and leverage autonomous agents that drive revenue, safety, and customer trust.

According to a recent study from *Innovatics*, nearly 87% of machine learning models fail during deployment. They never reach production. Manual monitoring leads to undetected model drift and risk. Also, only 54% of AI projects advance from pilot to production at best. MLOps tools and technologies address these failures. Enterprises are investing billions trying to solve the ML model deployment crisis.

MLOps is the standardised framework that unifies data science, DevOps, and governance to ensure models are not just built, but are reliable, compliant, and cost-effective at scale. It bridges the gap between the people who build the models and those who run the business. Its pillars are:

Machine learning: Focused on model architecture, feature engineering, and hyperparameter tuning.

DevOps: Focused on CI/CD (continuous integration/continuous deployment), infrastructure-as-code, and system uptime.

Data engineering: This focuses on the ‘plumbing’ — it addresses ingesting, cleaning, and versioning the massive datasets required to train models.

According to *Grant View Research*,

the global MLOps market is projected to grow from US\$ 2.19 billion (2024) to US\$ 16.6 billion by 2030 at a CAGR of 40.5%. As per a report from *Innovatics*, MLOps spending in the US grew from nearly zero to over US\$ 2 billion in 2024, and is projected to reach US\$ 17-40 billion by 2030.

Core principles of MLOps

MLOps enables automation across the ML lifecycle, reducing errors and accelerating delivery, while aligning data scientists, ML engineers, data engineers, and DevOps teams around shared processes and tools.

Four continuous practices form the backbone of MLOps.

Continuous integration (CI) in MLOps: This automates the merging